Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/21/2005
PI Name:	Delp, Michael Ph.D.		
Project Title:	Arterial Remodeling and Functional Adaptations Induced by Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	None		
PI Email:	mdelp@fsu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	352-214-1195
Organization Name:	Florida State University		
PI Address 1:	College of Human Sciences		
PI Address 2:	242 Sandels Building, 120 Convocation Way		
PI Web Page:			
City:	Tallahassee	**State:**	FL
Zip Code:	32306-0001	**Congressional District:**	2
Comments:	Previous affiliations were University of Florida (mid-2007-June 2014), West Virginia University (mid-2005 to mid-2007), and Texas A&M University (1995 to mid-2005).		
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	03/15/2000	**End Date:**	02/14/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle		
Flight Assignment:	NOTE: PI moved to West Va. U from TX A&M in Sept 2005; grant transferred to WVU as NNA06CB60G (per A. Chu 7/06) STS-107		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC2-1166		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Tail suspension or hindlimb unloading of rats has been used as a ground-based model of microgravity because tail suspension, like actual microgravity, 1) induces a headward shift of fluids and 2) diminishes the contractile activity of hindlimb skeletal muscles that are normally involved in maintaining posture. It is now evident that the shift of fluid toward the head and the unloading of postural muscles alter the mechanical forces exerted on arteries, the vessels responsible for regulating blood flow and arterial blood pressure. For example, the smooth muscle cells in the small arteries of hindlimb skeletal muscle atrophies with tail suspension, and results in a diminished ability of these vessels to constrict. The compromised ability to constrict reduces the ability of the rats to maintain normal blood pressure, and consequently, these animals are susceptible to orthostatic hypotension (an abnormal reduction in arterial pressure when the body position of the rat is moved from a horizontal position to a 70o vertical head-up position). The purpose of the present study is to determine whether the fluid shifts and reduced activity of postural muscles that are induced by microgravity similarly alter rodent arterial vessel structure and function. In other words, does space flight result in a diminished ability of arteries to constrict, and if so, is this the result of smooth muscle atrophy. This work will provide potentially important information about the mechanisms underlying the reduced fitness and the inability to properly regulate blood pressure of astronauts returning to Earth. The purpose of the present study is to determine whether the fluid shifts and reduced activity of postural muscles that are induced by microgravity similarly alter rodent arterial vessel structure and function. In other words, does space flight result in a diminished ability of arteries to constrict, and if so, is this the result of smooth muscle atrophy. This work will provide potentially important information about the mechanisms underlying the reduced fitness and the inability to properly regulate blood pressure of astronauts returning to Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Orthostatic intolerance is a common health problem for Americans, particularly the elderly. Therefore, this work will provide potentially important information about the mechanisms underlying orthostatic intolerance and the reduced fitness as it pertains to vascular biology on Earth.
Task Progress:	A. List of Pre-, In-, Post-flight Anomalies Pre-flight: no anomalies In-flight: there were initial problems with maintaining temperature in SPACEHAB Post-flight: animals were not recovered B. Completeness/Quality of Data: Data were unable to be collected C. Summary of Results: None to report D. Photographs: None to report E. Status of Data Analysis: Not applicable F. Final Investigation Results: None to report due to the Columbia Shuttle accident G. Conclusion: None to report
Bibliography Type:	Description: (Last Updated: 07/09/2025)
Abstracts for Journals and Proceedings	Prisby RD, Delp MD. "Vasoconstrictor responses of the femoral principal nutrient artery." Experimental Biology Conference, April 2004. FASEB J. 2004 Apr;18: A656. , Apr-2004
Articles in Peer-reviewed Journals	Ray CA, Vasques M, Miller TA, Wilkerson MK, Delp MD. "Effects of short-term microgravity and long-term hindlimb unloading on rat cardiac mass and function." J Appl Physiol. 2001 Sep;91(3):1207-13. PMID: 11509517 , Jan-2003